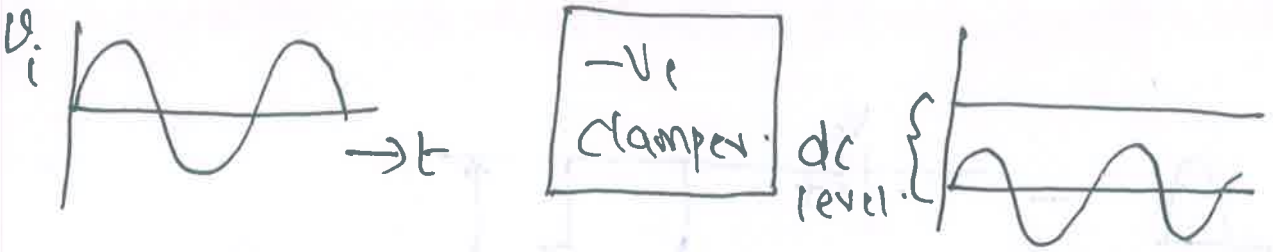
Clampers.

Clampers are circuits that add dc level to the circuits. Clamper circuits consists of input voltage source, Resistor, capacitor, Battery, load resistor and diode.

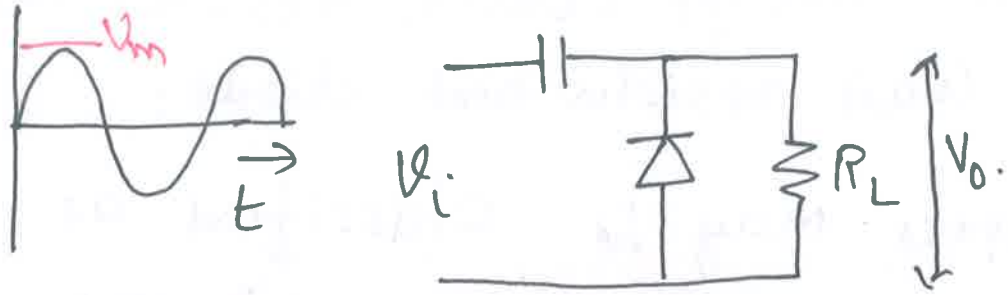
Clampers may be classified as positive clampers, negative clampers, positive biased clampers and negative biased clampers. Following are positive and negative clampers:



The dc level shift occuring in clampers is due to the presence of capacitor. The magnitude of dc level shift depends on the capability of the capacitor storage capacity.

For further change in the dc level can be effected by the use of battery.

Positive clamper:

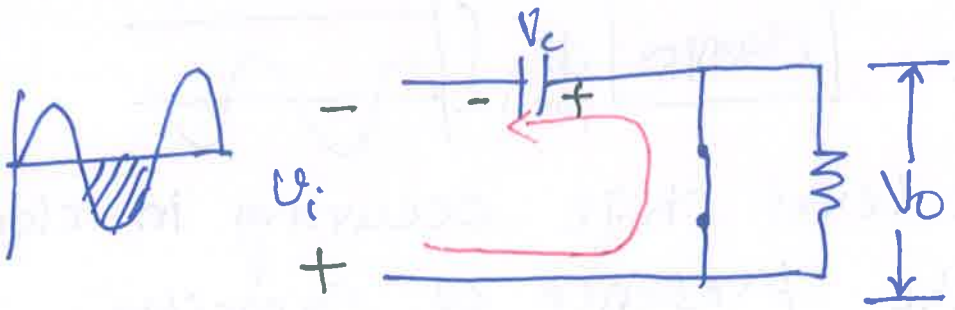


Steps for analysis:

1] Always start the analysis from the input cycle that forward biases (FB) the diode.

In the above circuit the -ve cycle FB the diode and hence start the analysis from -ve cycle of i/p voltage.

PS: Diode is considered ideal.



2] Observe the following:

(a) polarity of i/p voltage: It's -/+ since the analysis pertains to negative cycle.

(b) Polarity of Capacitor : Since the Current flows as shown in the diagram, Capacitor gets charged to −/+ [−‖+]
Since the Current flows from ~~Right~~ Left to ~~Left~~ Right [L←R].

3] During this cycle Calculate the output Voltage, V_o and the Capacitor charge voltage V_c.

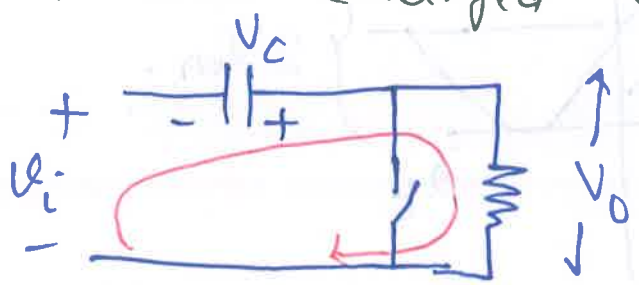
$V_o = 0$ [Since Current flows through the SC path of diode and 'V' across SC = 0].

Applying KVL in Loop shown in fig.

$+v_i - V_c = 0$

$V_c = +v_i = V_m$ [v_i V_m.]

4] Now, during the positive cycle, diode is RB due to both i/p Voltage and due to Capacitor charged Voltage.



★ Again observe the polarity of v_i and V_c.

5] The current flows as shown in fig. Now, it is important to determine the output voltage, V_o.

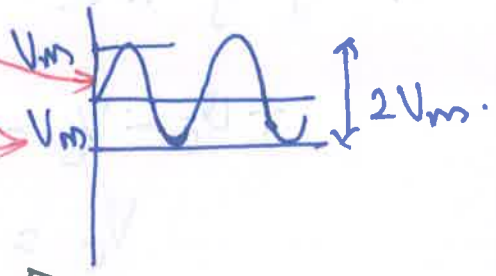
Remember that, capacitor has charged to a matm. value of V_m in the previous cycle.

PS: Time constant $\tau = RC$ is assumed to be very large so that the capacitor takes long time to discharge.

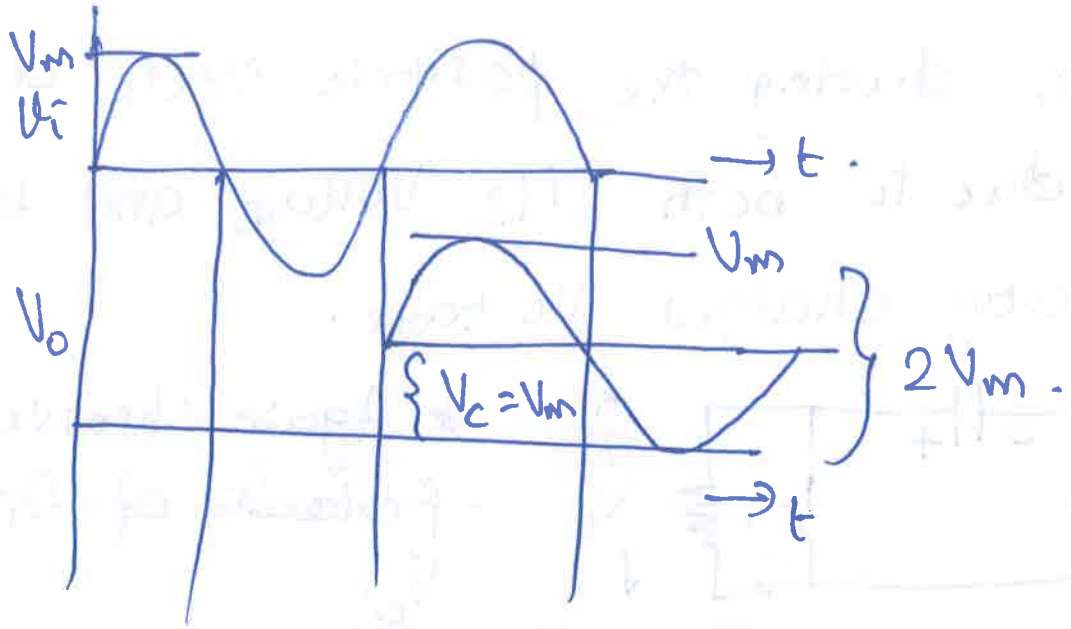
Applying KVL in the circuit.

$$+v_i + V_c - V_o = 0.$$

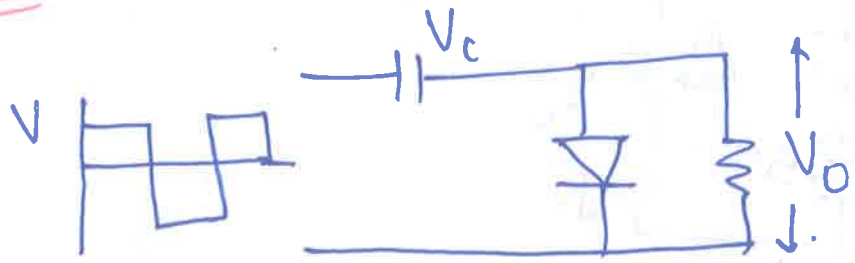
$$V_o = v_i + V_m.$$



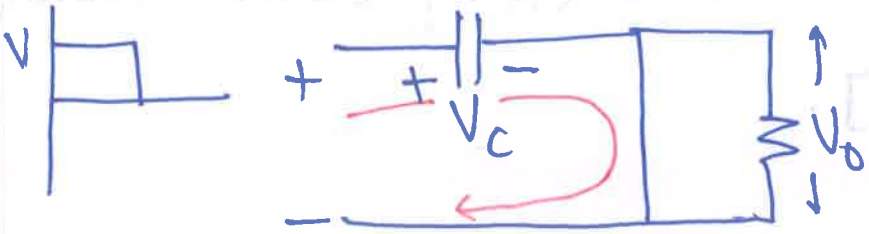
⑥ Combine V_o in step 3 and 5.



eg:2 Draw $V_0 = ?$



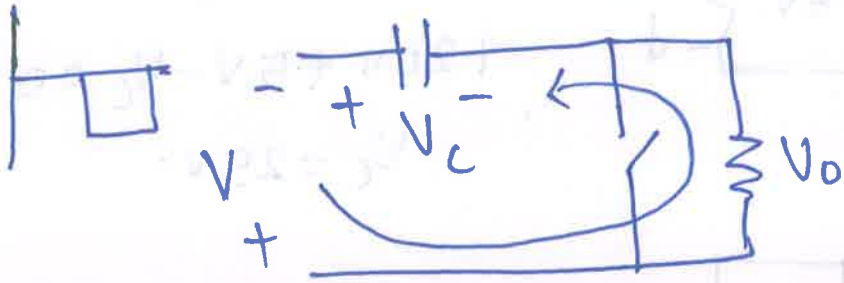
Step 1:



$V_0 = 0$

$V_c = V$

Step 2:

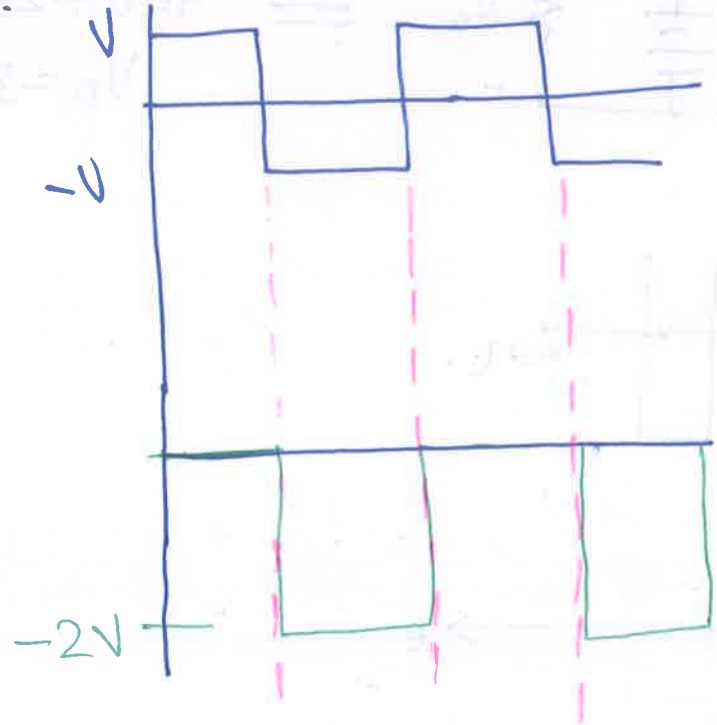


$+V + V_0 + V_c = 0$

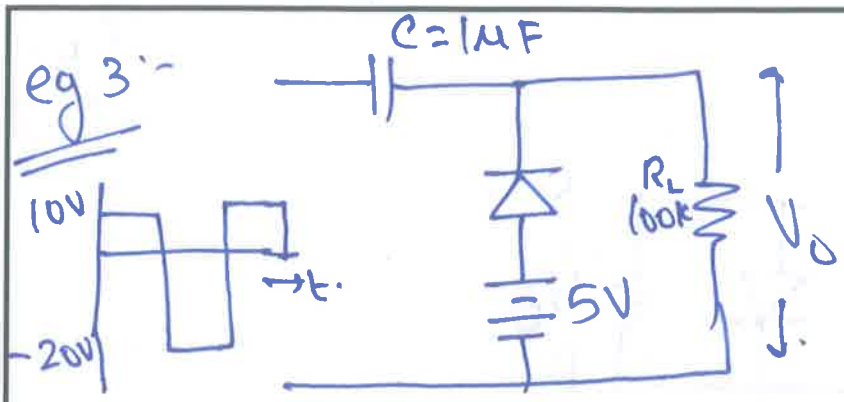
$V_0 = -V - V$

$V_0 = -2V$

Step 3:

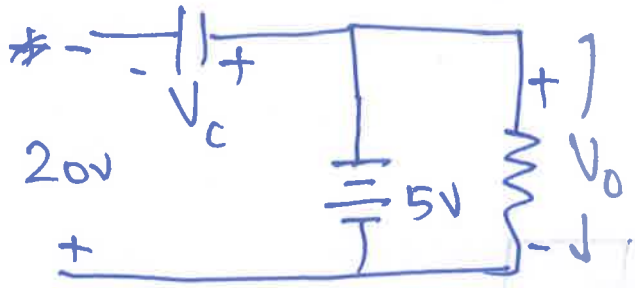


eg 3:-



1] Analyse from ~~+ve~~ -ve cycle. [Do not consider battery biasing].

2]

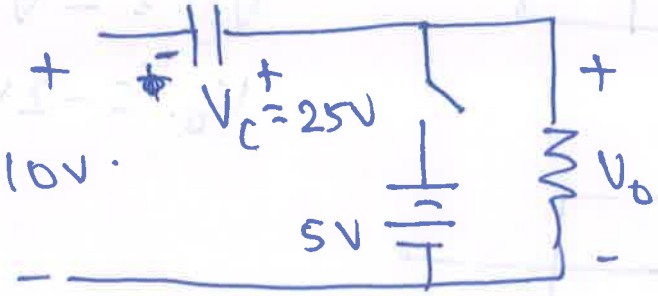


$V_o = 5V$.

$+20V + 5V - V_c = 0$.

$V_c = 25V$.

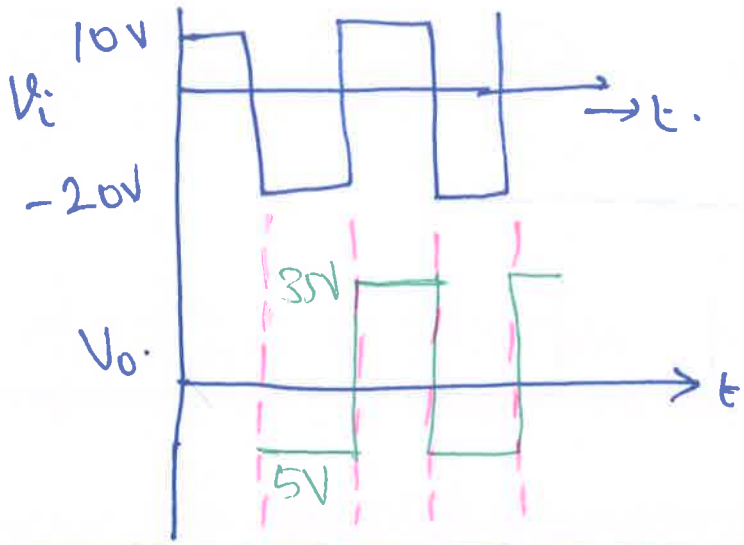
3]



$\underline{\underline{V_o}}$: $+10 + 25 - V_o = 0$.

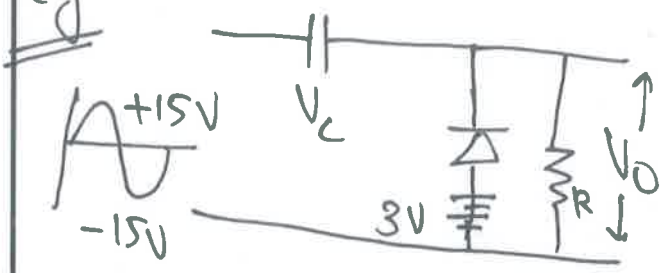
$V_o = 35V$.

4]

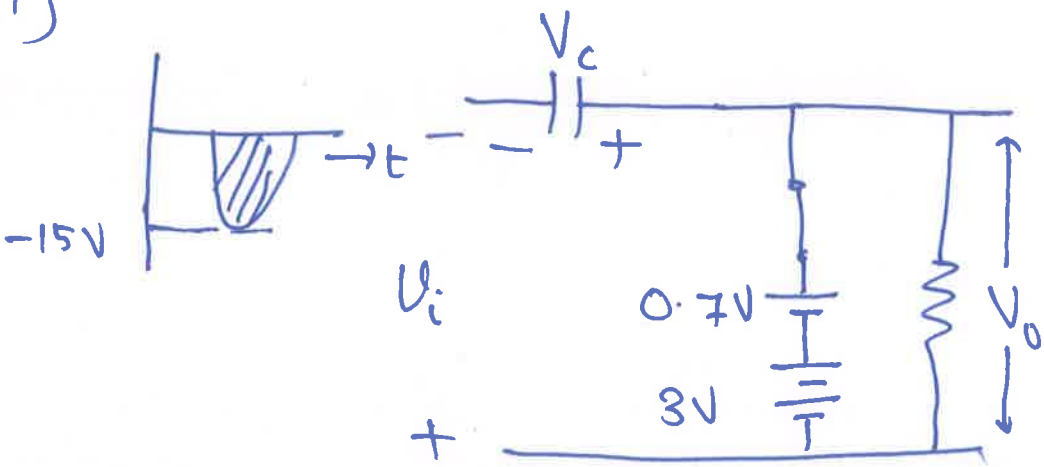


Now considering a practical diode with ac i/p
Sinusoidal i/p.

eg:



1]



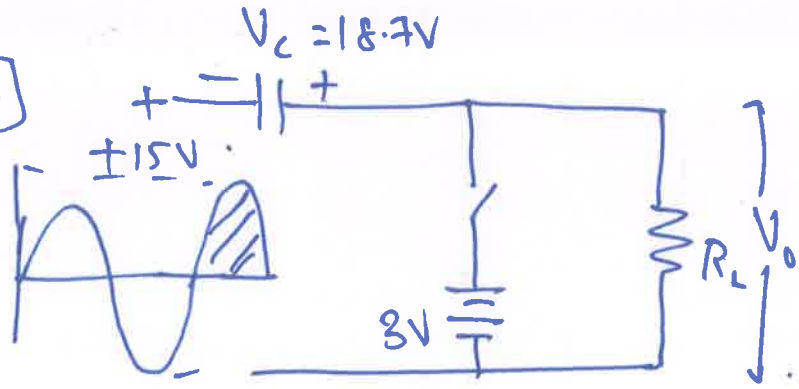
$V_0 = 3.7V.$

$+v_i + 3V + 0.7V - V_c = 0$

$V_c = 3.7 + 15$

$V_c = 18.7V.$

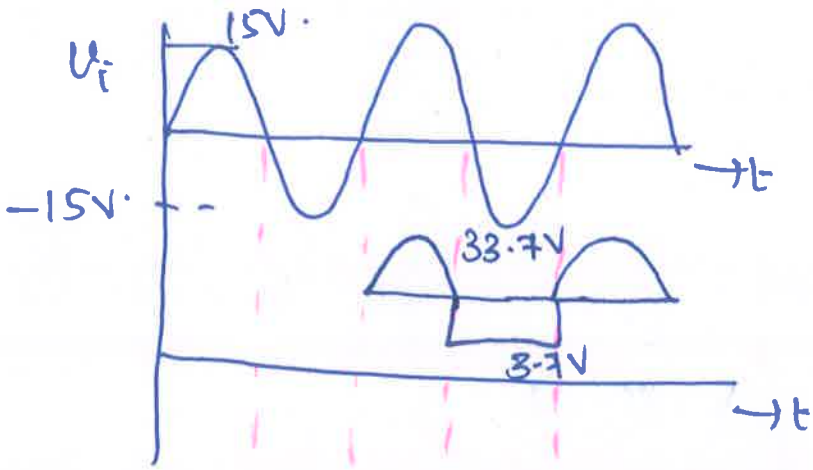
2]



$v_i + V_c - V_0 = 0.$

$V_0 = v_i + 18.7V.$

$V_m(+ve) = 15 + 18.7$

$= 33.7V.$

3]